

“Online Examination Portal”

A Project Report Submitted in the partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

In

DEPARTMENT OF COMPUTER SCIENCE ENGINEERING

&

DEPARTMENT OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

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Declaration

The Project Report entitled **“Online Examination Portal”** is a record of Bonafide work of **ARROLLA SHASHANK REDDY - 2520090139 , POTNURU ANUDEEP – 2520030087** submitted in partial fulfillment for the award of B. Tech in Computer Science and Engineering (or) Computer Science and Information Technology to the K L University. The results embodied in this report have not been copied from any other departments/University/Institute.

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CERTIFICATE

This is certify that the mini project based report entitled **“Online Examination Portal”** is a bonafide work done and submitted by **ARROLLA SHASHANK REDDY - 2520090139 , POTNURU ANUDEEP – 2520030087** in partial fulfillment of the requirements for the award of the degree of **BACHELOR OF TECHNOLOGY** in Department of Computer Science Engineering, K L (Deemed to be University), during the academic year **2024-2025**.

Signature of the Guide

Signature of the Course Coordinator

Signature of the HOD

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Abstract

The Online Examination Platform is a web-based system designed to make conducting exams easier, faster, and more efficient. Traditional exam methods are time-consuming, require manual grading, and can be difficult to manage, especially in online or large-scale learning environments. There is also a growing need for systems that can provide quick results while maintaining fairness and reliability.

This project offers a simple solution by allowing teachers to create exams with multiple-choice and essay questions through an easy-to-use exam builder. The platform automatically grades objective questions, saving time and reducing human error, while subjective answers can be reviewed manually. Features like a timer and basic anti-cheating mechanisms, such as tab-switch detection, help ensure that exams are conducted fairly.

Additionally, the system provides clear result displays and performance analytics to help track student progress and identify learning gaps. Results can also be exported as PDF reports for easy sharing and record-keeping. An admin panel enables smooth management of users, exams, and settings, making the entire process organized and user-friendly.

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CHAPTER -1 INTRODUCTION

1.1 Background of the Project

Education has always been the cornerstone of societal progress, and the ways in which academic institutions assess knowledge have evolved dramatically over time. For much of the twentieth century, examinations were conducted entirely on paper — question papers were printed, distributed to students in halls, answers were written manually, and scores were calculated by human examiners. While this model served its purpose, it came with significant logistical burdens: paper procurement, secure printing, physical invigilation, manual correction, and lengthy result declaration cycles.

The advent of the internet and personal computing introduced the concept of computer-based tests (CBT). In the early 2000s, institutions began experimenting with standalone software installed on examination center machines. These systems eliminated paper and enabled instant scoring for objective questions, but they were tied to physical locations and required specialized infrastructure. Students still had to travel to designated test centers, and administrators had to maintain dedicated hardware.

The next major shift came with the widespread adoption of broadband internet and modern web browsers. Web-based examination portals emerged as a superior alternative — students could potentially take tests from any internet-connected device, and administrators could manage exams from a central dashboard accessible from anywhere. Institutions worldwide, from GATE and JEE in India to SAT and GMAT internationally, migrated toward online examination systems. The COVID-19 pandemic in 2020-2021 accelerated this transition dramatically, forcing universities and schools to conduct assessments entirely online, sometimes within days of campus closures.

However, most existing enterprise-grade examination platforms are complex, expensive, and often overkill for smaller educational units such as departmental assessments, coaching institutes, or individual faculty-managed evaluations. Such platforms require dedicated backend servers, database administration, DevOps expertise, and recurring hosting costs. For a final-year B.Tech project or a departmental pilot, these requirements are prohibitively complex.

This project, ExamPortal, was conceived in this context. Recognizing that a modern web browser itself can serve as a capable runtime environment — with `localStorage` providing client-side persistence, React providing reactive UI rendering, and the browser timer APIs enabling countdown functionality — the project explores how far a purely frontend architecture can go in delivering a complete, professional-grade examination system. The result is a fully functional portal that requires no backend, no database server, and no cloud infrastructure while offering features comparable to commercial products.

The academic background for this project draws from multiple domains of computer science: Human-Computer Interaction for UI/UX design, Software Engineering for modular and maintainable architecture, Web Technologies for component-based development with React, and Data Management for structured localStorage schema design. Together, these disciplines converge into a system that is both technically rigorous and practically useful.

1.2 Problem Statement

The traditional approach to conducting examinations in educational institutions involves significant manual effort and carries multiple systemic weaknesses. Faculty members spend considerable time designing question papers, printing them securely, distributing them, collecting answer sheets, manually evaluating responses, and recording results. Each of these steps is susceptible to human error, and the entire cycle from examination to result declaration can take weeks.

Even institutions that have partially migrated to digital assessment often rely on generic tools — such as Google Forms or Microsoft Teams quizzes — that lack the specialized features required for formal examinations: countdown timers, anti-retake mechanisms, detailed per-question analytics, structured result reports, and separate administrative dashboards. These ad hoc solutions also make it difficult to maintain a history of student performance across multiple assessments.

The core problems that this project addresses are:

- The absence of a lightweight, self-contained examination system that requires no server setup or database administration.
- The lack of role-based access in generic tools, which forces administrators and students to use the same interface without appropriate privilege separation.
- The unavailability of automated, instant scoring and detailed answer review immediately after submission.
- The difficulty in generating formatted performance reports that can be exported or printed without additional software.
- The absence of a persistent theme preference in educational tools, leading to a poor user experience in low-light environments.
- The lack of an activity log that records important portal events for the administrator's reference.

ExamPortal directly addresses each of these pain points through a purpose-built, role-aware, fully client-side web application.

1.3 Scope of the Project

The scope of ExamPortal is deliberately defined to maximize utility within the constraints of a frontend-only architecture. The system is designed for use within a single browser session and across sessions on the same device, leveraging localStorage for state persistence. The following aspects are within scope:

1. User Authentication: Registration and login for student users, and credential-based login for an administrator account. Sessions persist across browser refreshes.
2. Exam Management: The administrator can create, edit, and delete examinations, configuring the title, description, time duration, and passing percentage for each.
3. Question Management: The administrator can add, edit, and delete multiple-choice questions (each with four options and a correct answer) for any exam.
4. Student Examination: Students can browse available exams, read instructions, take timed exams with question navigation and mark-for-review functionality, and submit answers.
5. Automated Scoring: Upon submission (manual or timer-triggered), the system instantly calculates the score, percentage, pass/fail status, and per-question correctness.
6. Results and Review: Students can view all past results and review each question with the answer they selected versus the correct answer.
7. Admin Analytics: The administrator dashboard displays system-wide metrics including total students, exams, attempts, average scores, and per-exam performance bars.
8. Report Export: Results can be exported to CSV or printed as a formatted HTML report from both the student result view and the admin reports page.
9. Theme Management: Users can switch between dark and light themes, with the preference persisted in localStorage.
10. Activity Logging: The system records major events (logins, exam creation, submission, etc.) in a time-stamped activity feed.

The following aspects are explicitly outside the current scope: real-time multi-user synchronization, third-party authentication (OAuth), online payment gateways, video proctoring, essay-type or subjective questions, and cloud hosting with a backend API.

1.4 Objectives of the Project

The primary objectives of this project are:

1. To design and implement a complete, role-based online examination portal using only frontend technologies, demonstrating the capability of modern browser APIs as a data persistence layer.
2. To develop a clean, modular React application that separates concerns using Context API providers for authentication, exam sessions, and theming.
3. To build an intuitive and responsive UI using Tailwind CSS with custom design tokens, supporting both dark and light modes, and rendering correctly on desktop, tablet, and mobile viewports.
4. To implement a real-time countdown timer system using JavaScript intervals within the React Context, with automatic submission on timer expiry.
5. To create a robust localStorage service layer that acts as a structured data store with typed operations for users, exams, questions, submissions, results, and logs.
6. To provide an administrator with comprehensive CRUD operations for exams and questions, a student management view with per-student analytics, a reports dashboard with date and exam filters, and settings for data management.
7. To enable students to review their submitted answers with correct/incorrect highlighting, export their results to CSV, and print formatted PDF-style reports.
8. To implement client-side route protection using ProtectedRoute components that verify authentication state and user role before allowing access to administrative or student panels.

1.5 Importance of the Application

The importance of ExamPortal extends beyond its technical implementation. From an educational standpoint, the application removes barriers that traditionally prevent smaller institutions or individual educators from adopting digital assessment. A faculty member can deploy this application on any static hosting service — GitHub Pages, Netlify, Vercel — in minutes, with zero cost and zero server maintenance. Students can access it from any device with a modern browser.

From a pedagogical perspective, instant feedback after exam submission is a well-documented tool for improving learning outcomes. Research in cognitive science consistently shows that learners who receive immediate, detailed feedback — knowing not just their score but exactly which questions they got wrong and why — retain information better and identify their weak areas more effectively. ExamPortal's answer review feature delivers exactly this kind of formative feedback.

From a data management perspective, the structured localStorage schema employed in this project demonstrates disciplined data modeling even without a traditional database. Each entity type (users, exams, questions, submissions, results, logs) has its own namespace key, and all operations are encapsulated in a dedicated service module, ensuring that the rest of the application interacts with data through a clean API rather than raw localStorage calls scattered throughout the codebase.

From an engineering perspective, the project demonstrates that the React Context API, when used thoughtfully with proper separation of concerns, can replace much of what a Redux store would provide, keeping the dependency footprint minimal. The entire application has only three runtime dependencies: React, React DOM, and React Router — a remarkably lean stack for a feature-rich product.

1.6 Target Users / Audience

ExamPortal is designed to serve two distinct categories of users:

Administrators (Faculty / Institution Staff):

The administrator role is intended for teaching faculty, exam coordinators, or department heads who are responsible for creating and managing assessments. The single admin account (username: admin, password: admin123) provides access to the full suite of management tools. Administrators are expected to have basic digital literacy — the ability to use a web browser and fill forms — but no technical knowledge is required.

Students:

Students are the primary end-users of the examination interface. They register using their name, username, email, and password, and thereafter access a personalized dashboard showing available exams and their performance history. Students are expected to be familiar with online interfaces and comfortable navigating web applications. The system is designed to be self-explanatory, requiring minimal instruction for first-time users.

Secondary audiences include academic institutions evaluating lightweight examination tools for pilot deployment, software engineering educators demonstrating frontend architecture patterns to students, and developers studying how to build full-featured applications with minimal dependencies

CHAPTER -2 SYSTEM REQUIREMENTS

2.1 Hardware Requirements

Since ExamPortal is a purely frontend web application that runs inside a browser, the hardware requirements are minimal. No dedicated server hardware is needed for deployment on a static hosting service. The following table outlines the recommended hardware specifications for both development and end-user usage:

Component	Minimum Specification	Recommended Specification
Processor	Dual-core CPU @ 1.6 GHz	Quad-core CPU @ 2.5 GHz or higher
RAM	2 GB	4 GB or more
Storage	500 MB free space	1 GB free space (for project files and Node.js)
Display	1024 × 768 resolution	1920 × 1080 (Full HD) or higher
Network	Not required (after initial load)	Broadband (for static hosting deployment)
Input Devices	Keyboard and Mouse / Touchscreen	Standard keyboard, mouse, or touch input

For student devices accessing the deployed application, only a device with a modern browser is required. The application has been designed to be responsive and functional on smartphones and tablets as well, making it accessible to students who may not have access to a desktop computer.

2.2 Software Requirements

The software requirements are categorized into development environment requirements and end-user runtime requirements:

Development Environment:

Software	Version Used	Purpose
Node.js	v18.x or higher	JavaScript runtime for development tooling and npm
npm (Node Package Manager)	v9.x or higher	Dependency management and script execution
Vite	v8.0.12	Build tool and development server with HMR
React	v19.2.6	Core UI library

Software	Version Used	Purpose
React DOM	v19.2.6	DOM rendering for React
React Router DOM	v7.16.0	Client-side routing and navigation
Tailwind CSS	v3.4.19	Utility-first CSS framework
PostCSS	v8.5.15	CSS transformation pipeline
Autoprefixer	v10.5.0	Automatic CSS vendor prefix addition
ESLint	v10.3.0	Static code analysis and linting
Git	v2.x or higher	Version control

End-User Runtime:

Browser	Minimum Version	LocalStorage Support
Google Chrome	80 or later	Yes (5–10 MB per origin)
Mozilla Firefox	75 or later	Yes
Microsoft Edge (Chromium)	80 or later	Yes
Apple Safari	13 or later	Yes
Opera	67 or later	Yes

No additional plugins, extensions, or software installations are required on the student's device. The application loads all required fonts from Google Fonts CDN on first load and caches them thereafter.

2.3 Development Tools and Frameworks

The following development tools were used throughout the project lifecycle:

Visual Studio Code:

The primary Integrated Development Environment (IDE) used for writing and editing source code. VS Code's React and Tailwind CSS extensions provided IntelliSense, code completion, and syntax highlighting, significantly accelerating development.

Vite (Build Tool):

Vite replaced the traditional Create React App (CRA) toolchain. Vite uses native ES modules in development, delivering near-instant Hot Module Replacement (HMR). Production builds are generated using Rollup, producing optimized, tree-shaken bundles. The vite.config.js configuration uses the @vitejs/plugin-react plugin for React's fast refresh integration.

Tailwind CSS CLI + PostCSS:

Tailwind CSS was configured via `tailwind.config.js` with custom design tokens for the ExamPortal color system (accent cyan #00E5FF, gold #FFD060, dark surface shades, and light mode variants). PostCSS processes the Tailwind directives and Autoprefixer adds vendor prefixes for cross-browser compatibility.

ESLint:

ESLint was configured with `eslint-plugin-react-hooks` to enforce the Rules of Hooks, and `eslint-plugin-react-refresh` to ensure components are compatible with Vite's fast refresh. This helped maintain code quality and catch potential issues early.

Google Fonts:

Three typefaces are loaded from Google Fonts CDN: Syne (font-display) for headings, DM Sans (font-body) for body text, and JetBrains Mono (font-mono) for statistical data and code-like values. These are referenced in the custom Tailwind `fontFamily` configuration.

Git and GitHub:

Version control was managed using Git, with the repository hosted on GitHub. Regular commits tracked the development progress, and the `.gitignore` file ensured `node_modules` and build artifacts were not committed.

CHAPTER-3 TECHNOLOGY STACK

ExamPortal is built on a carefully selected, deliberately minimal technology stack. Every dependency was chosen for a specific reason, and the overall footprint was kept as small as possible to avoid unnecessary complexity. The following sections describe each technology in detail.

3.1 Frontend

React 19:

React is a declarative, component-based JavaScript library for building user interfaces. Version 19 (the latest stable release at the time of development) introduced further improvements to the concurrent rendering engine. In ExamPortal, React is used for all UI rendering. The application is structured as a tree of functional components, each responsible for a specific portion of the interface. React's virtual DOM diffing ensures that only changed portions of the UI are re-rendered, resulting in smooth and efficient user interactions — particularly important during live exam sessions where the timer updates every second.

React Router DOM v7:

React Router DOM v7 provides the client-side routing system. The application uses BrowserRouter with nested Routes for a hierarchical URL structure (/admin/dashboard, /student/exams/:id, etc.). Route protection is implemented via a custom ProtectedRoute component that checks authentication state and user role from AuthContext. Dynamic routing (useParams) is used in the ExamInterface to load the correct exam by ID from the URL.

Tailwind CSS v3:

Tailwind CSS is a utility-first CSS framework that allows styling to be applied directly in JSX through class names. ExamPortal extends the default Tailwind configuration with a custom design system: a surface color palette (surface-900 through surface-400) for dark mode backgrounds, an accent color (#00E5FF cyan) and a gold color (#FFD060) for highlights and accents, a light mode palette for card and border colors, and custom animation keyframes for fade-in, slide-up, slide-in, and glow effects. Dark mode is controlled via the 'class' strategy, toggled by adding/removing the 'dark' class on the <html> element.

Context API (React built-in):

State management is handled entirely by React's built-in Context API, without any external state management library. Three context providers wrap the application: AuthContext for authentication state, ExamContext for active exam session management, and ThemeContext for theme preference. This approach eliminates the need for Redux or Zustand while keeping global state cleanly separated by concern.

3.2 Backend

ExamPortal follows a frontend-only architecture. There is no backend server, no REST API, and no server-side logic. All business logic — authentication verification, score calculation, result storage, session management — executes entirely within the user's browser.

This architectural decision was intentional. It ensures that the application can be hosted on any static file server (GitHub Pages, Netlify, Vercel, Cloudflare Pages) at zero cost, with no ongoing server maintenance. It also means the application works entirely offline once loaded in the browser.

In a future production deployment, a backend could be introduced using Node.js with Express.js or a serverless function platform (AWS Lambda, Vercel Functions) with a database like PostgreSQL or MongoDB. The localStorage service layer in the current implementation is designed with this migration in mind — the function signatures could be adapted to call an API rather than localStorage with minimal changes to the rest of the codebase.

3.3 Database / Storage

The application uses the browser's localStorage API as its data persistence layer. localStorage is a synchronous key-value store available in all modern browsers, with a typical storage quota of 5–10 MB per origin. Data is persisted across browser sessions (unlike sessionStorage) and is scoped to the origin (protocol + domain + port).

ExamPortal defines eight distinct storage keys, each namespaced with the 'ep_' prefix to avoid

conflicts with other applications on the same origin:

Storage Key	Data Type	Contents
ep_users	JSON Array	Registered student objects with id, name, username, email, role, joinedAt
ep_exams	JSON Array	Exam objects with id, title, description, duration (minutes), passingPercentage, createdAt
ep_questions	JSON Array	Question objects with id, examId, question text, optionA–D, correctAnswer, marks
ep_submissions	JSON Array	In-progress exam attempts with studentId, examId, answers, markedForReview, startedAt, status
ep_results	JSON Array	Completed result objects with scores, percentages, per-question answer details
ep_session	JSON Object	Currently logged-in user object (persists across refresh)
ep_theme	String	Current theme preference: 'dark' or 'light'
ep_activity_logs	JSON Array	Activity log entries with text, icon, and timestamp (max 50 entries, LIFO)

All `localStorage` interactions are routed through a dedicated service module (`localStorageService.js`) that exposes typed read/write functions. Raw `localStorage` calls do not appear anywhere else in the codebase, ensuring that the storage layer is centralized and easily replaceable.

3.4 Version Control

Git was used as the version control system throughout the development lifecycle. The repository was initialized at project creation, and commits were made at logical checkpoints: after completing each feature module, after fixing a bug, or after a significant refactor. The `.gitignore` file (generated by Vite) excludes `node_modules/`, `dist/`, `.env` files, and OS-specific metadata files.

The project can be cloned from the repository and run locally with a simple npm install followed by npm run dev.

3.5 APIs / External Libraries

ExamPortal intentionally avoids external API dependencies to ensure offline functionality and zero ongoing costs. However, it does leverage several browser-native APIs:

localStorage API:

For persistent client-side data storage across all entity types as described in section 3.3.

Window.setInterval / clearInterval:

Used within ExamContext for the countdown timer. A one-second interval is established when an exam begins and is cleared on submission, navigation away, or component unmount. The interval callback decrements the remaining seconds and triggers automatic submission when the timer reaches zero.

Blob API and URL.createObjectURL:

Used in csvExport.js to generate downloadable CSV files client-side. A Blob is created from the CSV string, an object URL is created from the Blob, a temporary anchor element clicks the URL, and the URL is immediately revoked to free memory.

Window.open / Document.write:

Used in pdfExport.js to generate print-ready reports. A new browser window is opened, and a fully styled HTML page is written to its document. The window's onload event triggers window.print(), opening the browser's native print dialog where the user can save as PDF or send to a physical printer.

Google Fonts CDN:

Three font families (Syne, DM Sans, JetBrains Mono) are loaded from fonts.googleapis.com via a <link> tag in index.html. These fonts are cached by the browser after the first load.

CHAPTER -4 SYSTEM ARCHITECTURE

4.1 High-Level Architecture

ExamPortal follows a Single-Page Application (SPA) architecture. The entire application is compiled into a set of static files (HTML, CSS, JavaScript bundles, and assets) by Vite during the build process. These files are served to the browser once, and all subsequent navigation and data operations happen client-side without any additional server requests.

The high-level architecture can be visualized as three conceptual layers:

Presentation Layer (React Components):

All UI rendering is handled by React functional components organized into pages, layouts, and reusable UI components. Components consume data from the Context layer through custom hooks and dispatch user actions that result in storage layer operations.

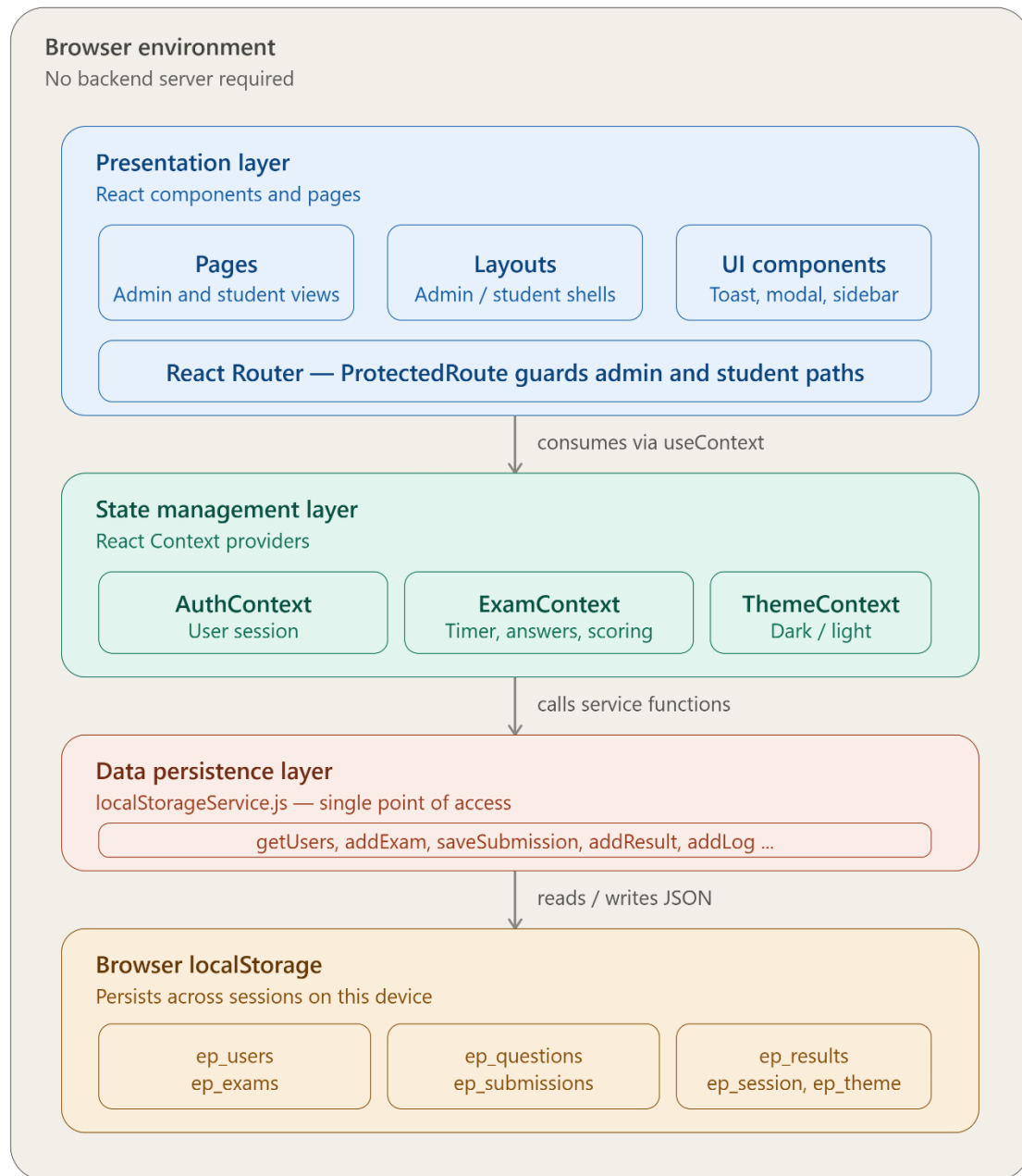
State Management Layer (React Context API):

Three Context providers maintain global application state: AuthContext manages the current user session, ExamContext manages the active exam attempt and timer, and ThemeContext manages the UI theme. These contexts act as the application's "state bus" — any component anywhere in the tree can subscribe to any context without prop drilling.

Data Persistence Layer (localStorage Service):

The localStorageService.js module provides all data access functions. It is the only part of the codebase that interacts with localStorage directly. All reads and writes go through this module, which handles JSON serialization/deserialization and error recovery.

4.2 Architecture Diagram (ASCII)



4.3 Layer-wise Explanation

Entry Point:

The browser loads `index.html`, which contains a single `<div id='root'>` and a `<script type='module'>` tag pointing to `src/main.jsx`. React mounts the `App` component into the root div.

Provider Hierarchy:

`App.jsx` wraps the entire application in four nested providers in order: `ThemeProvider` (outermost), `AuthProvider`, `ExamProvider`, and `ToastProvider` (innermost). This ordering ensures

that authentication state is available to the exam context, and that all contexts are available to the toast notification system. The AppInner function component, defined inside App.jsx, is wrapped in BrowserRouter and contains the route definitions.

Route Guarding:

The ProtectedRoute component wraps all authenticated route sections. On render, it reads the user state from AuthContext. If no user is found, it redirects to /login. If the user exists but their role does not match the required role prop (admin or student), they are redirected to their appropriate dashboard.

Data Seeding:

On first load, the AppInner useEffect calls seedDemoData() from localStorageService.js. This function checks for the ep_seeded key in localStorage — if absent, it populates the store with three sample exams (Data Structures & Algorithms, Database Management Systems, Web Development Basics) and five questions each, and sets ep_seeded to prevent re-seeding on subsequent loads.

4.4 Deployment Architecture

The deployment architecture is entirely static. After running npm run build, Vite generates a /dist directory containing:

- index.html — the single HTML entry point with hashed asset references
- assets/ — chunked and minified JavaScript bundles, CSS files, and static assets

This /dist directory can be uploaded to any static hosting provider. No Node.js runtime, no database, and no environment variables are required in the hosting environment. The application works immediately on deployment. For local development, npm run dev starts Vite's development server on http://localhost:5173 with HMR enabled.

CHAPTER -5 DESIGN

5.1 Use Case Diagram (Text Representation)

The system has two actors: Administrator and Student. The following use cases are supported:

Administrator Use Cases:

1. Login with admin credentials
2. Create, edit, and delete exams
3. Add, edit, and delete multiple-choice questions for any exam
4. View all registered students with search and performance stats
5. View system-wide analytics on the admin dashboard
6. Filter and export reports by date range and exam
7. Print admin performance report
8. Export all results to CSV
9. Configure portal settings (theme, portal name, danger actions)
10. Reset demo data, clear exams, results, or student accounts
11. Logout

Student Use Cases:

12. Register a new account
13. Login with registered credentials
14. View personalized dashboard with stats and available exams
15. Browse all available (not yet attempted) exams
16. Read exam instructions before starting
17. Start a timed exam and navigate between questions
18. Select an answer for any question
19. Mark a question for review and return to it later
20. Submit the exam manually or have it auto-submit on timer expiry
21. View immediate results with score, percentage, and pass/fail status
22. Review each question with selected vs. correct answer highlighted
23. Export individual result to CSV
24. Print individual result as a formatted report
25. View history of all past results
26. Update profile information
27. Toggle dark/light theme
28. Logout

5.2 Storage Schema (ER-style)

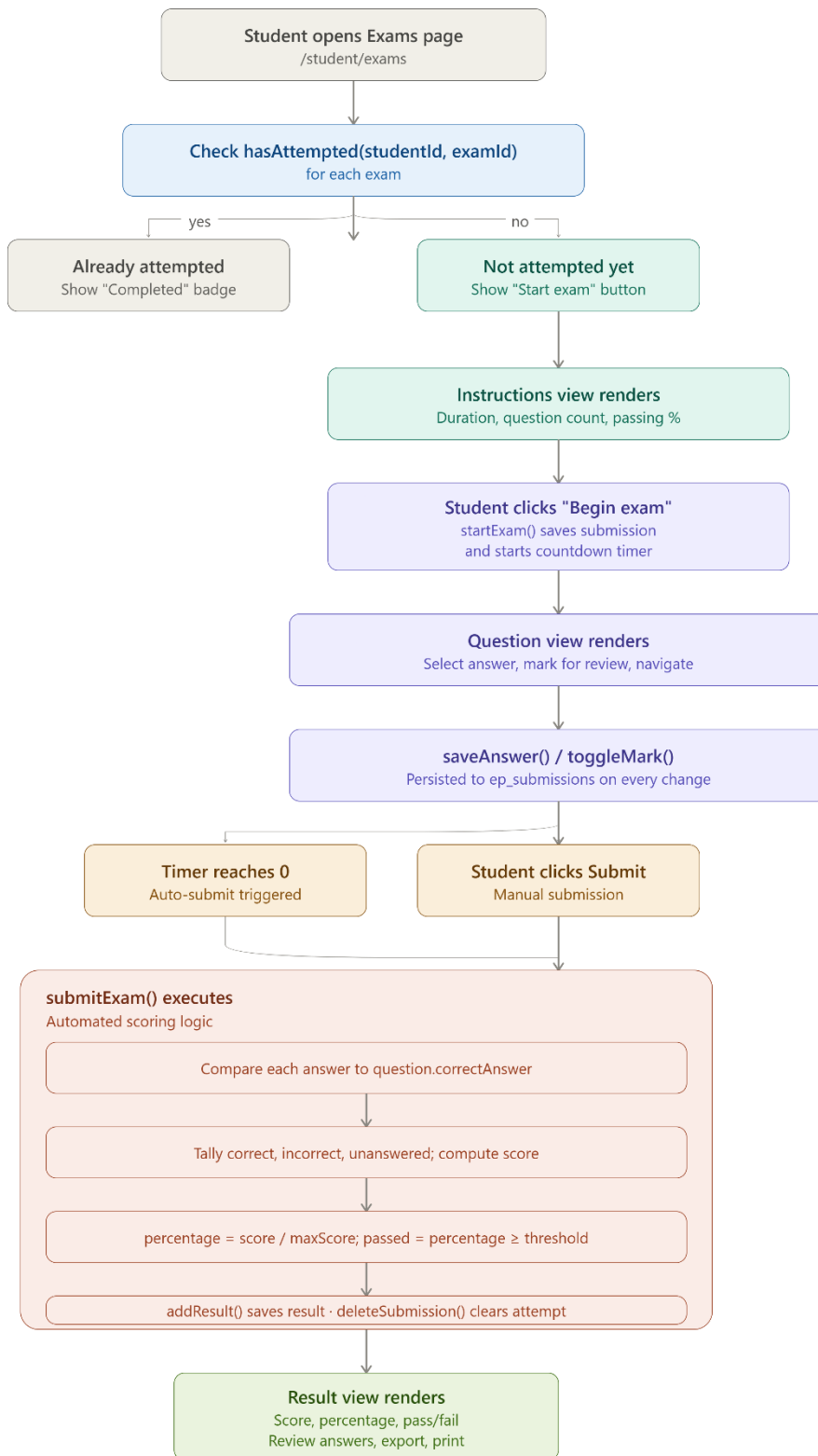
Since the application uses localStorage rather than a relational database, the following describes the logical data schema as it would be represented in a relational model:

Entity	Key Fields	Relationships
User	id (PK), name, username, email, password, role, joinedAt	One user → many Results, many Submissions
Exam	id (PK), title, description, duration, passingPercentage, createdAt	One exam → many Questions, many Results, many Submissions
Question	id (PK), examId (FK→Exam), question, optionA–D, correctAnswer, marks	Many questions belong to one Exam
Submission	studentId (FK→User), examId (FK→Exam), answers (object), markedForReview, startedAt, status	In-progress attempt linking User and Exam
Result	id (PK), studentId (FK→User), examId (FK→Exam), score, maxScore, percentage, passed, answers (array)	Completed attempt; one per student per exam
ActivityLog	id (PK), text, icon, time	Independent entity; records system events

In the localStorage implementation, each entity type is stored as a JSON-serialized array under its respective key. Foreign key relationships are maintained by storing ID references (e.g., result.studentId matches user.id), and joins are performed in JavaScript at the service layer when cross-entity data is needed.

5.3 Exam Attempt Flowchart

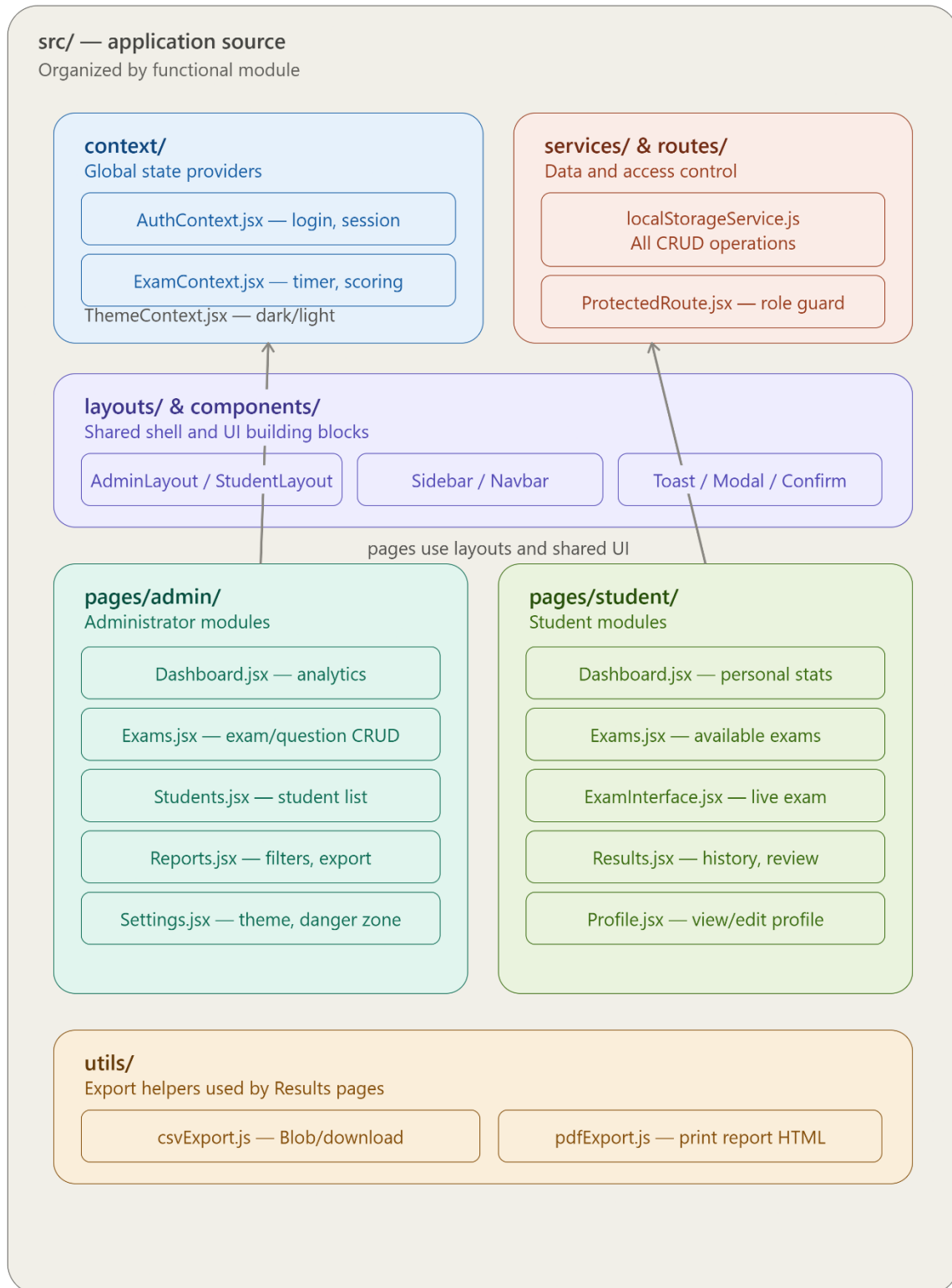
The following flowchart describes the complete lifecycle of a student's exam attempt:



CHAPTER -6 IMPLEMENTATION

Section 6.1 – Module-Wise Implementation

The project source code is organized into a modular structure under the /src directory:



6.2 Authentication Module

Authentication is handled entirely by `AuthContext.jsx`. On initialization, the provider reads the `ep_session` key from `localStorage`. If a valid session JSON is found, the user state is set immediately — this allows sessions to survive browser refreshes without re-login.

The login function handles two credential types. For the admin account, credentials are compared against a hardcoded constant (ADMIN object with `id`, `username`, `password`, `role`). For student accounts, the function calls `getUsers()` to retrieve the student list from `ep_users` and searches for a matching username and password combination. In both cases, the password field is stripped from the safe object before saving the session and setting the user state, ensuring that the password is never stored in the active session.

The register function validates uniqueness on both username and email before calling `addUser()` to append the new user to `ep_users`. The newly registered user must then log in separately — there is no auto-login after registration.

The `updateProfile` function merges updated data into the current user state and syncs both the session (`ep_session`) and the users list (`ep_users`). The logout function clears the user state and removes the `ep_session` key.

6.3 Routing

Routing is configured in `App.jsx` using React Router v7's `BrowserRouter` and `Route` components. The route structure is:

Route Path	Component	Access
/	Landing.jsx	Public
/login	Login.jsx	Public
/register	Register.jsx	Public
/admin	ProtectedRoute (role=admin) → AdminLayout	Admin only
/admin/dashboard	AdminDashboard.jsx	Admin only
/admin/exams	AdminExams.jsx	Admin only
/admin/students	AdminStudents.jsx	Admin only
/admin/reports	AdminReports.jsx	Admin only
/admin/settings	AdminSettings.jsx	Admin only
/student	ProtectedRoute (role=student) → StudentLayout	Student only

Route Path	Component	Access
/student/dashboard	StudentDashboard.jsx	Student only
/student/exams	StudentExams.jsx	Student only
/student/exams/:id	ExamInterface.jsx	Student only
/student/results	StudentResults.jsx	Student only
/student/profile	StudentProfile.jsx	Student only
*	Navigate to /	Public (catch-all)

Index routes redirect automatically: /admin → /admin/dashboard and /student → /student/dashboard. Any unmatched URL redirects to the landing page.

6.4 State Management

State management in ExamPortal is structured across three React Context providers, each managing a distinct concern:

AuthContext:

Manages: user (the current logged-in user object or null), isAdmin (boolean), isStudent (boolean). Exposes: login(), logout(), register(), updateProfile(). The user state is the single source of truth for authentication throughout the application.

ExamContext:

Manages: activeAttempt (the current exam session object containing exam, questions, answers dict, markedForReview array, startedAt, studentId), timeLeft (remaining seconds as an integer). Uses useRef (timerRef, attemptRef) to maintain mutable references that are accessible inside interval callbacks without stale closure issues. Exposes: startExam(), saveAnswer(), toggleMark(), submitExam(), cancelAttempt(). The use of useRef for the attempt alongside useState is a deliberate pattern to avoid stale closures in the timer's callback while still allowing React to re-render when state changes.

ThemeContext:

Manages: theme ('dark' | 'light'). Exposes: toggleTheme(), isDark (boolean). The useEffect hook that runs on theme change applies/removes 'dark' and 'light' classes on document.documentElement, which is the mechanism Tailwind's class-based dark mode uses.

6.5 Storage Logic

The localStorageService.js module follows a consistent pattern: a private read(key, fallback) function wraps JSON.parse with error recovery, and a private write(key, data) function wraps JSON.stringify. All exported functions use these primitives. Key design decisions include:

- Immutable updates: array mutations are performed by spreading existing arrays and adding/filtering elements, never by mutating in place.
- Atomic writes: each write operation replaces the entire key value atomically, avoiding partial update states.
- Error recovery: the read function returns the fallback value on parse failure, preventing crashes if localStorage contains malformed data.
- Activity logging: the addLog function prepends new entries and trims the array to 50 entries, preventing unbounded growth.

6.6 Component Interaction

The ExamInterface.jsx page is the most complex component in the application, combining multiple context consumers and localStorage service calls. When a student navigates to /student/exams/:id, the component:

29. Reads the exam ID from useParams()
30. Calls getExams() and finds the matching exam
31. Calls getExamQuestions(examId) to load questions
32. Calls hasAttempted(studentId, examId) — if true, renders a 'Already Completed' message
33. If not attempted, renders the Instructions component
34. On [Begin Exam] click, calls startExam(exam, questions, user.id) from ExamContext
35. Renders the QuestionView with the active attempt from ExamContext
36. Each answer selection calls saveAnswer(questionId, answer)
37. Each mark toggle calls toggleMark(questionId)
38. On submit, calls submitExam() which returns the result object
39. The component stores the returned result in local state and renders ResultView

CHAPTER 7- FEATURES

This chapter describes every feature of ExamPortal from both the user's perspective (what they see and experience) and the technical perspective (how it is implemented).

7.1 Role-Based Authentication

User Perspective:

When a user visits the portal, they are greeted by a landing page with options to Sign In or Get Started (Register). The registration form collects full name, username, email, and password. Upon successful registration, the user is directed to log in. The login form accepts username and password. After login, users are automatically directed to their role-appropriate dashboard — admins to /admin/dashboard and students to /student/dashboard. If a user tries to access a URL outside their role (e.g., a student accessing /admin/dashboard), they are redirected automatically.

Technical Perspective:

The ProtectedRoute component reads the user state from AuthContext. Role enforcement is implemented using conditional Navigate components: !user → redirect to /login, user.role !== required role → redirect to appropriate dashboard. Session persistence is achieved by reading the ep_session key from localStorage in AuthContext's useState initializer function, which runs synchronously before the first render.

7.2 Exam Management (Admin)

User Perspective:

The admin navigates to the Exam Management page and sees a list of all exams. A 'Create Exam' button opens a modal form where they enter the exam title, description, duration (in minutes), and passing percentage. Once created, the exam appears in the list. The admin can click any exam to select it and view its questions in a panel alongside the list. Individual questions can be added, edited, or deleted. Each question form collects the question text, four options, the correct answer, and the marks value.

Technical Perspective:

The AdminExams.jsx page maintains local component state for the exam list, selected exam, question list, modal visibility, and form data. CRUD operations call the corresponding service functions: addExam(), updateExam(), deleteExam(), addQuestion(), updateQuestion(), deleteQuestion(). Each successful operation calls addLog() to record the event in the activity feed. The delete operation is guarded by a ConfirmDialog component that requires explicit confirmation before proceeding. When an exam is deleted, its associated questions are also deleted by deleteExam().

7.3 Live Timed Exam Interface

User Perspective:

After reading instructions, the student clicks Begin Exam. The interface shows the current question with four radio-button options, a countdown timer in the top bar, and a question navigator sidebar showing numbered buttons for all questions. Answered questions appear in a different color, and questions marked for review show a distinctive indicator. The student can click any question number to jump to it. At any point, they can click Submit to finish the exam.

Technical Perspective:

The countdown timer runs via `setInterval(1000ms)` established in `ExamContext`. Each tick decrements `timeLeft` state by 1. When `timeLeft` reaches 1, the interval clears and `submitExam()` is called automatically. Answer selections are persisted to `ep_submissions` on every change via `saveAnswer()`, providing crash recovery — if the browser is accidentally closed and reopened during an exam, the in-progress submission exists in `localStorage` and could be loaded (though automatic resumption is currently not implemented in the UI).

7.4 Automated Scoring and Result Generation

User Perspective:

Immediately after submission, the student sees a result card showing whether they passed or failed, their score (e.g., 8/10), percentage, and counts of correct, incorrect, and skipped questions. A progress bar visualizes the percentage against the passing threshold. They can then choose to review answers question by question, export results to CSV, or print a formatted report.

Technical Perspective:

The `submitExam()` function in `ExamContext` iterates over all questions in the attempt, compares the stored answer (`answers[q.id]`) to the question's `correctAnswer`, and increments the score by `q.marks` for each correct answer. It also counts correct, incorrect, and unanswered questions. The percentage is calculated as `Math.round((score / maxScore) * 100)`, and passed is determined by comparing this percentage to `exam.passingPercentage`. The result object, including a per-question answer detail array, is persisted to `ep_results`.

7.5 Answer Review

User Perspective:

After seeing the summary result, the student can click 'Review Answers' to expand a section showing every question. Correct answers are highlighted in green, wrong answers in red, and skipped questions in grey. The student's selected option and the correct option are both clearly marked.

Technical Perspective:

The per-question `answerDetails` array stored in the result object contains: `questionId`, question text, selected answer, correct answer, and `isCorrect` boolean. The `ResultView` component iterates this array and applies conditional CSS classes based on the `isCorrect` and `selected` fields. This data is computed once at submission time and stored permanently, ensuring the review is always available even after navigating away and returning.

7.6 Admin Analytics Dashboard

User Perspective:

The admin dashboard displays four stat cards: Total Students, Total Exams, Total Attempts, and Average Score with pass rate. Below these are two mini-metric cards for overall pass rate and average score with animated progress bars. A 'Recent Registrations' table shows the last five students, and an Activity Feed shows the last ten log entries with timestamps. A per-exam performance section shows a breakdown bar for each exam.

Technical Perspective:

All dashboard data is computed live from `localStorage` reads on component mount. `getUsers()`, `getExams()`, `getResults()`, and `getLogs()` are called directly in the component body (not in a `useEffect` with loading state, since `localStorage` is synchronous). Average score is calculated as a reduce over all results' percentages.

Pass rate is the count of results where `passed === true` divided by total results.

7.7 Student Management

User Perspective:

The admin navigates to Student Management and sees a searchable table of all registered students with columns for name, username, attempt count, average score, and join date. Clicking a student row opens a detail panel showing their individual results for each exam they have attempted.

Technical Perspective:

The search filter is a client-side JavaScript filter on the students array using `String.toLowerCase().includes()`. The `getStudentStats` helper function filters results by `studentId` and computes per-student statistics on the fly.

7.8 Reports and Export

User Perspective:

The Reports page (admin only) allows filtering results by date range (All Time, Last 7 Days, Last 30 Days, Last 90 Days) and by specific exam. Aggregate stats update dynamically. Results are displayed in a table with student name, username, exam, score, percentage, and status. The admin can click Export CSV to download all filtered data or Print Report to open a formatted printable report.

Technical Perspective:

Date filtering is implemented using `useMemo`, which recalculates the filtered results array whenever the `dateFilter` or `examFilter` state changes. The cutoff date is computed as `Date.now() - days * 86400000` and compared against `result.submittedAt`. CSV export uses the `exportAdminCSV` utility, which calls

downloadCSV() to create a Blob and trigger a download. The print function calls printReport(buildAdminReport(stats, studentRows, examRows)), which generates styled HTML and triggers window.print().

7.9 Dark / Light Theme

User Perspective:

Both admin and student users can toggle between a dark theme (deep navy/slate palette with cyan accents) and a light theme (white cards, light grey backgrounds, dark text). The preference persists across browser refreshes and sessions. The toggle is available in the admin Settings page and also accessible from the sidebar for both user types.

Technical Perspective:

ThemeContext stores the theme preference and saves it to ep_theme in localStorage. A useEffect applies 'dark' or 'light' class to document.documentElement, which activates Tailwind's darkMode: 'class' system. Components that support both themes use conditional class strings based on the isDark boolean from useTheme(), applying either dark palette classes or light palette classes.

7.10 Toast Notification System

User Perspective:

Whenever a significant action is performed — creating an exam, deleting a question, submitting an exam, or encountering an error — a small notification toast appears in the bottom-right corner of the screen and auto-dismisses after a few seconds.

Technical Perspective:

Toast.jsx implements a ToastProvider context that maintains an array of active toast messages. The useToast() hook exposes a toast(message, type) function. Adding a toast appends an object with a unique ID, message, and type ('success', 'error', 'warning', 'info') to the array and sets a setTimeout to remove it. Toast messages are rendered as absolutely positioned, animated cards overlaid on the page content.

7.11 Student Profile

User Perspective:

The student can visit their Profile page to view their name, username, email, and join date. They can edit their full name and email directly on this page.

Technical Perspective:

Profile edits call updateProfile(data) in AuthContext, which merges the new data into the current user state, saves the updated object to ep_session, and if the user is a student, calls updateUser(id, data) to update the ep_users array.

CHAPTER -8 TESTING

Since ExamPortal is a purely frontend application with no backend API, formal automated testing frameworks such as Jest or Cypress were not integrated in this version. Testing was conducted through systematic manual testing covering functional scenarios, UI/UX behavior, and edge cases. All tests were performed in Google Chrome (latest version) and verified in Mozilla Firefox.

8.1 Functional Testing

Functional testing verified that every feature of the application behaves as specified. Each module was tested in isolation before integration testing was performed across modules.

8.2 Test Cases and Results

TC #	Module	Test Description	Expected Output	Result
TC-01	Auth	Register with new unique username and email	Account created; redirect to login	PASS
TC-02	Auth	Register with duplicate username	Error: 'Username already exists'	PASS
TC-03	Auth	Register with duplicate email	Error: 'Email already registered'	PASS
TC-04	Auth	Login with valid admin credentials (admin/admin123)	Redirect to /admin/dashboard	PASS
TC-05	Auth	Login with valid student credentials	Redirect to /student/dashboard	PASS
TC-06	Auth	Login with incorrect password	Error: 'Invalid username or password'	PASS
TC-07	Auth	Session persistence after browser refresh	User remains logged in	PASS
TC-08	Auth	Admin tries to access /student/dashboard	Redirect to /admin/dashboard	PASS
TC-09	Auth	Student tries to access /admin/dashboard	Redirect to /student/dashboard	PASS
TC-10	Auth	Unauthenticated user tries /admin/dashboard	Redirect to /login	PASS
TC-11	Exam Mgmt	Admin creates new exam with valid data	Exam appears in list; activity logged	PASS

TC #	Module	Test Description	Expected Output	Result
TC-12	Exam Mgmt	Admin creates exam with empty title	Toast error: 'Exam title is required'	PASS
TC-13	Exam Mgmt	Admin edits exam duration	Duration updated in list and localStorage	PASS
TC-14	Exam Mgmt	Admin deletes exam with questions	Exam and all its questions removed	PASS
TC-15	Question Mgmt	Admin adds question with all 4 options	Question saved and appears in list	PASS
TC-16	Question Mgmt	Admin adds question with missing option	Toast error: 'All 4 options are required'	PASS
TC-17	Exam Flow	Student views available exams	Only non-attempted exams shown with start button	PASS
TC-18	Exam Flow	Student starts exam and reads instructions	Instruction card shows duration, Q count, passing %	PASS
TC-19	Exam Flow	Student answers all questions and submits	Score calculated; result displayed immediately	PASS
TC-20	Exam Flow	Timer reaches zero during exam	Exam auto-submits with current answers	PASS
TC-21	Exam Flow	Student revisits an attempted exam	'Already Completed' message shown; no restart allowed	PASS
TC-22	Exam Flow	Student marks question for review	Question marked; indicator shown in navigator	PASS
TC-23	Scoring	All answers correct	Score = maxScore; 100%; PASS if threshold $\leq$ 100%	PASS
TC-24	Scoring	No answers selected	Score = 0; 0%; FAIL	PASS
TC-25	Scoring	Percentage equals passing threshold exactly	Result is PASS	PASS
TC-26	Scoring	Percentage one below threshold	Result is FAIL	PASS
TC-27	Reports	Admin export CSV with results present	CSV file downloads with correct headers and data	PASS

TC #	Module	Test Description	Expected Output	Result
TC-28	Reports	Admin print report	New window opens with styled report; print dialog appears	PASS
TC-29	Reports	Filter by Last 7 Days with no results in range	Empty table; stats show 0	PASS
TC-30	Theme	Toggle from dark to light	UI switches to light palette; ep_theme set to 'light'	PASS
TC-31	Theme	Theme persists after browser refresh	Same theme applied on reload	PASS
TC-32	Profile	Student updates name	Name updated in session and users list	PASS
TC-33	Settings	Admin resets demo data	Original 3 exams and 15 questions restored; results cleared	PASS

8.3 UI and Responsiveness Testing

Viewport	Device Class	Observations
1920 × 1080	Desktop (Full HD)	All layouts render at full width; sidebar visible at all times; 4-column stat grids display correctly
1280 × 800	Laptop	Full layout preserved; no horizontal scroll; all features accessible
768 × 1024	Tablet (portrait)	Sidebar collapses to mobile mode with hamburger toggle; 2-column grids activate; exam interface navigable via touch
375 × 812	Mobile (iPhone)	Single column layout; all buttons fully tappable; question navigator scrollable; timer clearly visible

CHAPTER -9 DEPLOYMENT

9.1 Installation Steps (Local Setup)

Follow these steps to run ExamPortal on a local development machine:

Step 1: Prerequisites

Ensure Node.js (v18 or higher) and npm (v9 or higher) are installed. Verify by running:

```
node -v  
npm -v
```

Step 2: Extract the Project

Extract the uploaded examportal.zip file to your preferred directory. The project root is the inner examportal/ folder containing package.json.

```
cd examportal
```

Step 3: Install Dependencies

```
npm install
```

This reads package.json and installs all dependencies into the node_modules/ directory. The installation typically takes 30-60 seconds on a standard connection.

Step 4: Start the Development Server

```
npm run dev
```

Vite starts the development server and outputs a localhost URL (default: http://localhost:5173). Open this URL in your browser.

Step 5: First Launch

On first load, the `seedDemoData()` function automatically populates `localStorage` with 3 sample exams and 15 questions. Demo admin credentials: username: admin, password: admin123.

9.2 Build Process

To create an optimized production build:

```
npm run build
```

Vite compiles, minifies, and bundles all source files into the `/dist` directory. The output includes:

- `index.html` with hashed asset links
- `assets/index-[hash].js` — the main JavaScript bundle (React, Router, and application code)
- `assets/index-[hash].css` — the compiled and purged Tailwind CSS

The build output can be previewed locally:

```
npm run preview
```

9.3 Hosting (Static Deployment)

The `/dist` directory is a self-contained static website. It can be deployed to any of the following services with no configuration changes:

Hosting Service	Deployment Method	Notes
Netlify	Drag-and-drop <code>/dist</code>	Free tier; custom domain; automatic

Hosting Service	Deployment Method	Notes
	folder or connect GitHub repo	HTTPS
Vercel	Import GitHub repo; set build command to npm run build	Free tier; excellent CDN performance
GitHub Pages	Push /dist contents to gh-pages branch	Free for public repos; custom domain support
Cloudflare Pages	Connect GitHub repo; build command: npm run build	Free tier; global CDN
Firebase Hosting	firebase deploy after firebase init	Free Spark plan available
Any Web Server	Copy /dist to web server document root	Apache, Nginx, IIS all supported

Since there is no backend, no environment variables need to be configured on the hosting platform. The deployment is complete as soon as the static files are served.

9.4 Environment Setup Note

There are no .env files or environment variables required for this application. The only external dependency at runtime is the Google Fonts CDN for loading Syne, DM Sans, and JetBrains Mono. If deploying in an environment without internet access, these fonts will fall back to the browser's default sans-serif font, and the application will remain fully functional.

CHAPTER -10 CHALLENGES & LIMITATIONS

10.1 Challenges Faced During Development

Challenge 1: Stale Closure in the Timer Callback

The most technically interesting challenge involved the countdown timer in ExamContext. When submitExam() was called from within the setInterval callback at timer expiry, it referenced a stale version of activeAttempt from the closure — the attempt object from when the interval was created, which did not include answers the student selected after the timer started.

Solution:

The solution was to maintain a useRef (attemptRef) alongside the useState (activeAttempt). Every time the student selected an answer via saveAnswer(), both the state (for re-rendering) and the ref (for the timer closure) were updated. The submitExam() function, when called automatically by the timer, reads from attemptRef.current rather than from the state variable, ensuring it always has the latest answers.

Challenge 2: Dark/Light Theme Consistency Across Components

With over 15 page-level components, ensuring consistent dark/light mode styling across every element was challenging. Early in development, some components rendered correctly in dark mode but showed poor contrast in light mode, or vice versa.

Solution:

A systematic approach was adopted: all conditional classes follow the pattern isDark ? 'dark-class' : 'light-class'. Custom color tokens in tailwind.config.js (surface-* for dark, light-* for light) provided a consistent palette. Additionally,

the custom Tailwind color aliases meant that a change to a color token automatically propagated to all components using it.

Challenge 3: Nested Route Protection

React Router v7's nested route structure required careful placement of `ProtectedRoute` to avoid wrapping individual leaf routes — which would cause multiple role checks and unnecessary re-renders.

Solution:

`ProtectedRoute` was applied at the layout level (`/admin` and `/student` paths), not at each individual page. This means the role check runs once when entering the admin or student section, and all nested routes inherit the protection.

Challenge 4: localStorage Size Limits

During development, it was noted that `localStorage` has a typical limit of 5-10 MB per origin. For a system with many students and many exam attempts, the stored data could theoretically approach this limit, especially if long questions or many results accumulate.

Solution:

The activity log is capped at 50 entries (the `addLog` function slices the array to 50 after every write). The Settings page provides danger zone actions to clear students, results, or exams, giving the admin control over data volume. Long-term, this is an argument for migrating to a backend database.

10.2 Current Limitations

Limitation	Description	Workaround / Note
No Multi-Device Sync	Data stored in localStorage is device- and browser-specific; different devices have different data	Future: backend API + database
Single Admin Account	Only one hardcoded admin account (admin/admin123) exists; credentials cannot be changed via the UI	Future: admin registration module
No Session Security	Passwords are stored in plain text in ep_users; no hashing is performed	Future: bcrypt hashing on a backend
No Exam Timer Resume	If the browser is closed mid-exam, the timer restarts from the original duration on next visit (not from where it paused)	The in-progress submission is preserved, but timer state is not
Subjective Questions Not Supported	Only 4-option MCQ format is supported; no essay, fill-in-the-blank, or numerical answer types	Future: extensible question type engine
No Proctoring	No webcam monitoring, tab-switch detection, or copy-paste prevention mechanisms exist	Future: browser Fullscreen API + blur events

Limitation	Description	Workaround / Note
No Negative Marking	All scoring is positive; no penalty for wrong answers	Future: configurable negative marks per exam
localStorage Storage Cap	5–10 MB browser limit may be reached for large-scale deployments	Future: backend database
No Email Verification	Student registration does not verify the provided email address	Future: OTP or email verification flow

CHAPTER -11 FUTURE ENHANCEMENTS

The current version of ExamPortal represents a solid, feature-complete foundation. The following enhancements are planned or proposed for future iterations:

11.1 Backend API Integration

The most impactful future enhancement is the introduction of a proper backend API. A Node.js application using the Express.js framework could expose RESTful endpoints for all the operations currently handled by `localStorageService.js`. The frontend service layer would be updated to call these endpoints via `fetch()` or `Axios`. The database would be PostgreSQL or MongoDB, providing ACID-compliant storage, multi-device access, and unlimited scalability. This change would also enable real-time features using WebSockets (Socket.IO) for live exam monitoring.

11.2 Secure Authentication

With a backend in place, authentication security can be dramatically improved. Passwords should be hashed using `bcrypt` before storage. JSON Web Tokens (JWT) should be used for session management, with short-lived access tokens and refresh token rotation. OAuth 2.0 integration with Google Sign-In would allow students to register and log in using their institutional Google accounts, eliminating password management entirely.

11.3 Multiple Question Types

Extending the question engine to support multiple question types would significantly increase the platform's academic utility:

- True/False questions with binary options
- Multiple-correct MCQs where more than one option may be correct

- Fill-in-the-blank with text matching (exact or fuzzy)
- Numerical answer questions with acceptable range tolerance

11.4 Exam Scheduling and Access Control

Admins should be able to schedule exams for specific time windows — defining start and end datetimes for each exam. Students attempting to access an exam outside its window would see an appropriate message. Additionally, exams could be locked to specific student groups or cohorts, preventing unauthorized access.

11.5 Online Proctoring Features

Basic anti-cheating mechanisms could be implemented using browser APIs: the Fullscreen API could force the exam into fullscreen mode, and the window blur event could detect and log tab switches. Webcam-based monitoring using the getUserMedia API could capture periodic snapshots. These features would significantly improve exam integrity for formal assessments.

11.6 Analytics and Reporting Enhancement

The current analytics are computed from raw arrays. With a backend, complex aggregate queries could be run at the database level. The admin dashboard could include interactive charts (using Recharts or Chart.js) showing performance trends over time, question-level difficulty analysis (percentage of students who got each question wrong), and comparison across cohorts or time periods.

11.7 Negative Marking and Partial Marking

Many competitive examinations use negative marking to discourage guessing. The scoring engine could be extended to support a configurable negativeMark value per exam (e.g., -0.25 per wrong answer). Similarly, partial marking for multiple-correct questions could award proportional marks based on the number of correct

options selected.

11.8 Progressive Web App (PWA)

The application could be converted to a Progressive Web App by adding a Web App Manifest and a Service Worker. This would allow students to install ExamPortal on their home screen, receive push notifications for upcoming exams, and take exams offline (for portions that don't require real-time data).

11.9 Accessibility Improvements

Full WCAG 2.1 AA compliance could be achieved by adding ARIA labels to all interactive elements, ensuring keyboard-only navigation through the exam interface, improving color contrast ratios for the light theme, and implementing focus management for modal dialogs and the question navigator.

12.1 Project Summary

ExamPortal is a complete, role-based online examination management system built entirely as a frontend web application. Starting from a recognition of the limitations of manual and ad hoc digital examination systems, the project set out to deliver a professional-grade solution using only the browser as its runtime environment. The result is a feature-rich application that handles user registration and authentication, exam and question management, live timed exam sessions, automated scoring, detailed result review, CSV and print-format report generation, system-wide analytics, and persistent theme preferences — all without a single line of backend server code.

12.2 Project Outcomes

The project successfully achieved all stated objectives. The application is functionally complete and has been tested across all major modules. It is deployable to any static hosting service in minutes, making it practically accessible to small educational institutions and individual educators. The admin dashboard provides meaningful insight into student performance, and the student interface delivers a smooth and professional exam-taking experience.

The architectural decision to use `localStorage` as the data persistence layer proved effective for the project's scope. The `localStorageService.js` abstraction ensures that this decision is encapsulated and replaceable, and the code demonstrates how disciplined data modeling can be applied even without a traditional database.

The project also demonstrates that React's Context API, when applied with clear separation of concerns across multiple focused providers (`AuthContext`,

ExamContext, ThemeContext), is a sufficient and maintainable alternative to external state management libraries for applications of this complexity. The three-context architecture delivered clean separation of authentication, exam session, and UI state without any Redux boilerplate.

12.3 Skills Learned During Development

The development of ExamPortal provided hands-on experience across a wide range of software engineering competencies:

React Advanced Patterns:

Practical application of the Context API with multiple providers, custom hooks (useAuth, useExam, useTheme, useToast), and the useRef-alongside-useState pattern for handling stale closures in timer callbacks.

React Router v7:

Nested routes, dynamic routes (useParams), index routes, programmatic navigation (useNavigate), and custom route guard components.

Tailwind CSS Design Systems:

Configuring a custom design token system within tailwind.config.js, implementing dark/light mode via the class strategy, and using utility classes to achieve complex responsive layouts without writing a single custom CSS class for structural purposes.

Frontend Architecture:

Designing a maintainable folder structure with clear separation between pages,

layouts, components, context, services, and utilities. Understanding how this structure scales and where the boundaries between layers should lie.

Client-Side Data Management:

Designing a structured localStorage schema, implementing CRUD operations on JSON arrays, and understanding the tradeoffs of client-side storage versus server-side databases.

Browser APIs:

Practical use of localStorage, setInterval/clearInterval, Blob API, URL.createObjectURL, window.open, and document.write for implementing core application features without third-party libraries.

Software Engineering Practices:

Writing modular, reusable code; implementing error handling and validation; designing for both happy paths and edge cases; and documenting code through consistent naming conventions and logical file organization.

In conclusion, ExamPortal stands as evidence that thoughtful software design, combined with a deep understanding of the browser's capabilities, can produce a genuinely useful product. It represents not just a B.Tech final-year submission but a practical tool — and a strong foundation for a production-grade examination platform.

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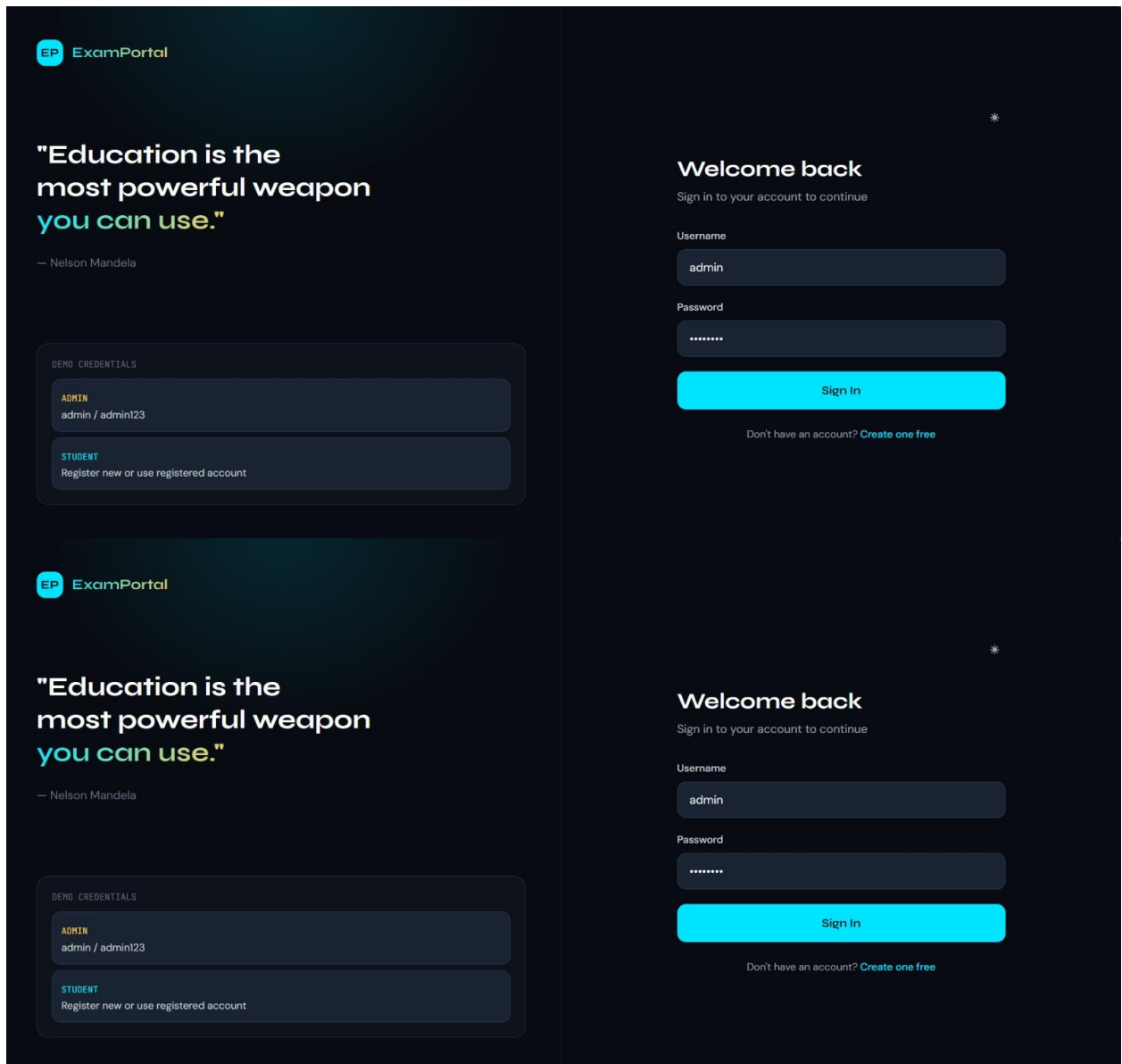
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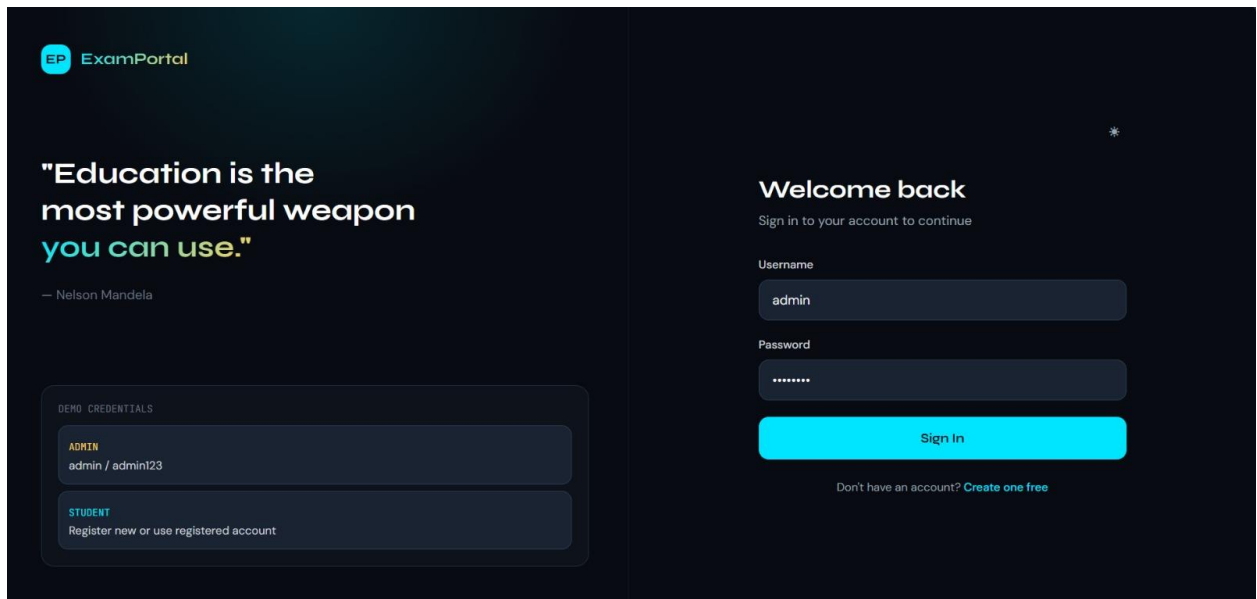
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CHAPTER -14 APPENDICES

Screenshots of the app





Sample code snippets

```
import { useAuth } from "../../context/AuthContext";
import { useTheme } from "../../context/ThemeContext";
import { getExams, getResults, getExamQuestions, hasAttempted } from "../../services/localStorageService";
import { Link } from "react-router-dom";

export default function StudentDashboard() {
  const { user } = useAuth();
  const { isDark } = useTheme();

  const exams = getExams();
  const allResults = getResults();
  const myResults = allResults.filter(r => r.studentId === user.id);

  const attempted = new Set(myResults.map(r => r.examId));
  const availableExams = exams.filter(e => !attempted.has(e.id));
  const avgScore = myResults.length ? Math.round(myResults.reduce((s, r) => s + r.percentage, 0) / myResults.length) : 0;
  const passed = myResults.filter(r => r.passed).length;

  const stats = [
    { label: "Available Exams", value: availableExams.length, icon: "📄", color: "text-accent bg-accent/10" },
    { label: "Completed", value: myResults.length, icon: "✅", color: "text-emerald-400 bg-emerald-400/10" },
    { label: "Average Score", value: myResults.length ? `${avgScore}%` : "-", icon: "📊", color: "text-gold bg-gold/10" },
    { label: "Passed", value: passed, icon: "🏆", color: "text-purple-400 bg-purple-400/10" },
  ];

  return (
    <div className="space-y-8 animate-fade-in">
      <div>
        <h1 className={`font-display font-bold text-3xl ${isDark ? "text-white" : "text-light-text"}`}>
          Hello, <span className="gradient-text">{user?.name?.split(" ")[0] || user?.username}</span> 🌟
        </h1>
        <p className={`font-body mt-1 ${isDark ? "text-gray-400" : "text-light-muted"}`}>Your exam dashboard – track progress and t
        </p>
      </div>
    </div>
  );
}
```

```

import { useTheme } from "../../context/ThemeContext";
import { getUsers, getExams, getResults, getLogs } from "../../services/localStorageService";

function StatCard({ label, value, sub, icon, color, isDark }) {
  return (
    <div className={`p-6 rounded-2xl border transition-all hover:-translate-y-1 duration-300 ${isDark ? "bg-surface-700 border-surf`
    <div className="flex items-start justify-between mb-4">
      <div className={`text-2xl p-2 rounded-xl ${isDark ? "bg-surface-600" : "bg-light-bg"}`}>{icon}</div>
      <span className={`font-mono text-xs px-2 py-1 rounded-full ${color}`}>{sub}</span>
    </div>
    <div className={`font-display font-bold text-3xl ${isDark ? "text-white" : "text-light-text"}`}>{value}</div>
    <div className={`font-body text-sm mt-1 ${isDark ? "text-gray-400" : "text-light-muted"}`}>{label}</div>
  </div>
  );
}

export default function AdminDashboard() {
  const { isDark } = useTheme();
  const users = getUsers();
  const exams = getExams();
  const results = getResults();
  const logs = getLogs();

  const avgScore = results.length ? Math.round(results.reduce((s, r) => s + r.percentage, 0) / results.length) : 0;
  const passRate = results.length ? Math.round(results.filter(r => r.passed).length / results.length * 100) : 0;
  const recentStudents = users.slice(-5).reverse();

  const stats = [
    { label: "Total Students", value: users.length, sub: "Registered", icon: "👤", color: "text-accent bg-accent/10" },
    { label: "Total Exams", value: exams.length, sub: "Created", icon: "📋", color: "text-gold bg-gold/10" },
    { label: "Total Attempts", value: results.length, sub: "Completed", icon: "📝", color: "text-purple-400 bg-purple-400/10" },
    { label: "Average Score", value: `${avgScore}%`, sub: `${passRate}% pass`, icon: "🏆", color: "text-emerald-400 bg-emerald-400/10" }
  ];
}

```

Appendix B — Installation Guide (Quick Reference)

```
# Prerequisites
node --version      # Must be v18.0.0 or higher
npm --version       # Must be v9.0.0 or higher

# Setup
cd examportal
npm install

# Development
npm run dev         # Starts at http://localhost:5173

# Production Build
npm run build       # Output in /dist
npm run preview     # Preview production build locally

# Demo Credentials
Admin:   username: admin | password: admin123
Student: Register a new account via /register
```

Appendix C — User Manual

For Administrators:

1. Open the portal URL and log in with username 'admin' and password 'admin123'.
2. Navigate to Exam Management. Click '+ Create Exam' to create a new exam with a title, description, duration, and passing percentage.
3. Click any exam in the list to select it. Use the question panel on the right to add MCQ questions with 4 options and the correct answer.
4. Navigate to Student Management to search and view registered student performance.
5. Navigate to Reports to filter and export performance data by date or exam.
6. Navigate to Settings to toggle the theme, adjust general settings, or use danger zone actions to reset data.

For Students:

7. Visit the portal URL. Click 'Get Started Free' to register with your name, username, email, and password.
8. Log in with your registered credentials.
9. Your Dashboard shows available exams and your overall statistics.
10. Navigate to Exams and click 'Start Exam' on any available exam to read instructions.
11. Click 'Begin Exam' to start the timer and begin answering questions.
12. Use the sidebar to navigate between questions. Click 'Mark for Review' to flag a question.
13. When ready, click 'Submit Exam' or wait for the timer to expire.
14. Review your result immediately. Export to CSV or print the report as needed.

15. Your complete exam history is available under the Results tab.

Geo Tag photos with guide

